Problem 1

Problem Set 2

(a)

Proof: we proved contrapositive:

if $a_1 <= a_3$, then there will be chain

there are possible cases

Case A: if $a_3 >= a_1$ & $a_3 > a_2$

it will make a chain of $a_1 < a_2 <= a_3$

Case B: if $a_3 >= a_1$ & $a_3 < a_2$

we get $a_1 < a_2$ & $a_1 <= a_3$ & $a_2 > a_3$

B.1 : $a_4 >= a_3$. this will create the chain of $a_1 <= a_3 <= a_4$

$a_2 > a_3 > a_4$

B.2 : $a_4 < a_3$. this "

∴ assuming $a_1 < a_2$, no chain $\Rightarrow a_3 < a_1$

(b) Proof: from the proof of (a), we know $a_1 < a_2$ & no 3 chain $\Rightarrow a_3 < a_1$

we have $a_1 < a_2$, $a_1 > a_3$

since there is no chain, $a_3 < a_2$. if not $a_3 >= a_2 \Rightarrow$ a chain of

$a_1 < a_2 <= a_3$

Lets think about a_4:

if $a_4 >= a_2$, we have the chain of $a_1 < a_2 <= a_4$

∴ therfore $a_4 < a_2$

if $a_4 <= a_3$, we have the chain of $a_2 > a_3 >= a_4$

from $a_3 < a_2$, $a_4 < a_2$ & $a_3 < a_4$

we get $a_3 < a_4 < a_2$

(c) if $a_5 \geq a_4$, we get $a_3 < a_4 <= a_5$

if $a_5 < a_4$ 5 " " $a2 > a4 > a5$

(d) proof: By Contradiction

Assume for the purpose of contrdiction that

there exist a sequence of five distinct integer $(a_1, a_2, a_3, a_4, a_5)$ that not form 3-chain

Snice all five integer are distinct, $a_1 < a_2$ (or) $a_1 > a_2$

let's assume $a_1 < a_2$ (if $a_1 > a_2$, the exact same logic will apply by symmetry or reversing the sequence)

By the proof of (a), we know $a_3 < a_1$

" (b), " $a_3 < a_4 < a_2$

" (c), " given $a_1 < a_2$ & $a_3 < a_4 < a_2$

any value of a_5 will create the chain

This is a contrdiction Therefor theory must be true.

Problem 2

problem 2

Proof: By induction, Let

$$P(n) = \left(\frac{n(n+1)}{2}\right)^2$$

Base case: $P(1) = 1^3 = \left(\frac{1(1+1)}{2}\right)^2 = 1$ ✓

Indutive steps for $n \geq 1$, prove $P(n) \Rightarrow P(n+1)$

for the purpose of induction, assume $P(n)$ is true.

$$1^3 + 2^3 + 3^3 + \cdots + n^3 + (n+1)^3$$

By I.H, $= \left[\frac{n(n+1)}{2}\right]^2 + (n+1)^3$

$$= \frac{n^2(n+1)^2}{4} + (n+1)^3$$

$$= (n+1)^2 \cdot \left(\frac{n^2+4n+4}{4}\right)$$

$$= (n+1)^2 \cdot \frac{(n+2)^2}{4}$$

$$= \left(\frac{(n+1)(n+2)}{2}\right)^2$$ ✓

$\forall n \geq 1$, $P(n+1) \Rightarrow P(n+1)$

By the principle of induction, $P(n)$ is true $\forall n \geq 1$.

Problem 3

problem 3

Thm: If fewer than n students in class are initially infected, the whole class will never be completely infected

Let: P be the perimeter of infected region as the number of exposed outer edges separating infected student from uninfected students

Lemma P(t) will only be equal to P(t-1) or less than P(t-1)

Pf: it need at least 2 adjacent students to infect a student.

Case A: if a student touch with 2 adjacent students
we got 2 extra parameter but touched parameter from 2 adjacent will also disapper
2-2=0

Cas B: if a student touch with 3,4 adjacent students,
just like case A, 2-3=-1
2-4=-2
lose the # of parameter

Pf for thm if less than n student are infected, we will use k (k<n) student
we get 4k parameter (4k<4n) of infected
to infected nxn student (which 4n parameter)
we need at least 4n parameter of infected student

By lemma, infected area parameter 4k will never increase
therfore it can't never completely infect the whole class.

Problem 4

problem 5

Flaw: The inductive step is invalid because it assume $a^n = 1$, which is not guoranteed by the inductive hypothesis when $n = 1$

$a^1 = a$

Problem 5

Problem 6

Pf: by strong induction

$P(n) ::= 3^n - 2^n$

Base Case

$P(0) = 3^0 - 2^0 = 0 = G_0$ ✓

Inductive step

$\forall n \geq 0$, show $P(1) \wedge P(2) \wedge \dots \wedge P(n) \rightarrow P(n+1)$

Assume $P(1) \wedge P(2) \wedge \dots \wedge P(n)$ are True for the purpose of strong induction.

$$G_{n+1} = 5G_n - 6G_{n-1}$$

By I.H $= 5(3^n - 2^n) - 6(3^{n-1} - 2^{n-1})$

$= 5 \cdot 3^n - 5 \cdot 2^n - 6\left(\frac{3^n}{3} - \frac{2^n}{2}\right)$

$= 5 \cdot 3^n - 5 \cdot 2^n - 6\left(\frac{2 \cdot 3^n - 3 \cdot 2^n}{6}\right)$

$= 5 \cdot 3^n - 5 \cdot 2^n - 2 \cdot 3^n + 3 \cdot 2^n$

$= 3 \cdot 3^n - 2 \cdot 2^n$

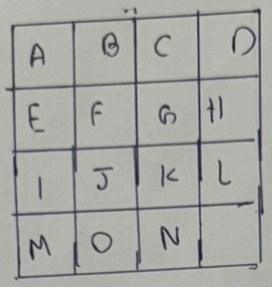
$= 3^{n+1} - 2^{n+1}$

$P(1) \wedge P(2) \wedge \dots \wedge P(n) \rightarrow P(n+1)$

By the principle of strong induction, $P(n)$ is true for $\forall n \geq 0$

Problem G

Theorem: No sequence of moves transforms the board below on the left into the board below on the right.



A	B	C	D
E	F	G	H
I	J	K	L
M	O	N	

A	B	C	D
E	F	G	H
I	J	K	L
M	N	O	

(a) Lemma 1: Row moves doesn't change the order

proof: we move a item from cell i to a blank cell $i \pm 1$, since no other cell are changed therefore order stay the same

(b) Lemma 2: column moves change the relative order of precisely 3 pairs of items.

proof:

we move a item from cell i to a blank cell $(i \pm 4)$, which change the order of the 3 intermediate steps $(i \pm 1, i \pm 2, i \pm 3)$.

(c) Lemma 3: Row move doesn't affect on the parity of # of inversion.

proof: by lemma 1

(d) Lemma 4: Column move always flip the parity

proof: there are four possible cases:

0 inversions in intermediate pair → # of inversion +3

1 " " " → " " ±1

2 " " " → " " -1

3 " " " → " " -3

(e) lemma 5: In every configuration reachable from the position shown below, the parity of the number is different from the pairty of row containing the blank square.

row 1	A	B	C	D
row 2	E	F	G	H
row 3	I	J	K	L
row 4	M	O	N	

Proof: by induction

P(n) := For any configuration reachable in n moves, the parity of # of inversion is different from the parity of the row containing blank space

Base Case:

P(0) = no move

the parity of # of inversion = odd

" of row containg blank space = even

P(0) = true

Inductive step: $n \geq 0$, show $P(n) \rightarrow P(n+1)$

Assume P(n) is true for the purpuse of induction

we know P(n) is True, P(n+1), there are two legal move:

Row move: by lemma 3, both the pairty of inversion & the parity of row stay the same

P(n+1) = True

Column move: the blank space move vertically ± 1, the parity of row flipped

By lemma 4, the parity of # of inversion also flipped

B.c both side are flipped, they reamain opposite.

P(n+1): True

$P(n) \rightarrow P(n+1)$

By the principle of induction P(n) is True $\forall n \geq 0$

Pf of thm the parity of # of inversion at desired state is even (0)

" of row " " is even (4)

By lemma 5, The desired state can't not be reached from initial stage □

Problem 7

Problem 7

Proof: By induction

$P(n) ::= Z_n < B_n$ after n reproducing

Base case:

$P(0) = 200 < 800$ ✓

Inductive step: $n \geq 0$, show $P(n) \rightarrow P(n+1)$

Assume $P(n)$ is true for the purpose of induction,

By I.H, we know $P(n)$ is True, at $P(n+1)$ there are two possible cases

Case A: $Z_n = B_n$

they will find each Z & B pair, creating all Z-B pair and maintaing Both Z_n & B_n number the same.

Case B: $Z_n < B_n$

first we will get Z_n pair of Z-B pairs

the remaining B-lings (B-Z) will from B-B pairs

(for odd, B-Z-1 will work the same)

we get (B-Z) B-ling and $\frac{(B-Z)}{2}$ of Z-ling

since $\frac{B-Z}{2}$ is half of difference of B-Z, adding that into Z_{n+1} will stay make less than B_{n+1}

$P(n) \rightarrow P(n+1)$ ✓

∴ By the principle of induction, $P(n)$ is T for $\forall n \geq 0$

$Z_n <= B_n$

$Z_n <= 2B_n$ (b.c B_n is positive population number)